Imprimitive Wreath Monodromy of Self-Composed Constraint Surfaces

The seam as imprimitivity system, the omega substrate,
and a forward/backward recovery asymmetry

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Draft — session results; certainty tiers marked inline.

June 27, 2026

Abstract

We study the branched-cover structure that arises when a constraint surface $F(D, S, P) = 0$, carrying a directive–substrate–product (DBP) role assignment, is composed with itself by identifying the product of one stage with the directive of the next. Eliminating the shared variable (the *seam*) yields a four-variable hypersurface $\widetilde{F}(D_1, S_1, S_2, P_2) = 0$ whose monodromy we compute directly. Three findings stand out. First, there is no single monodromy group: each role-output cover carries its own, and each is a *full wreath product that is imprimitive with respect to the seam.* For $F = D^{m_D} + DS + P^{m_P} - 3$ the product cover is $\mathrm{Sym}(m_P) \wr \mathrm{Sym}(m_P)$ and, at $m_P = 2$, the directive cover is $\mathrm{Sym}(m_D) \wr \mathrm{Sym}(m_D)$, with degree spectrum (m_D^2, m_D, m_P, m_P^2). Second, the second-stage substrate S_2 is an *omega role*: it controls the position of the downstream product branch while leaving the seam invariant ($\partial A/\partial S_2 = 0$, $\partial C/\partial S_2 = 1$), and looping it steers exactly one product branch pair at a time. Third, iterated composition produces a sharp recovery asymmetry: reconstructing the root directive over n stages is degree m_D^n while producing the final product is degree m_P^n. We also bound a tempting "wave/phase" reading: the abelianisation of the directive group is $(\mathbb{Z}/2)^2$, so its $U(1)$ characters are signs and the higher (cubic) structure is non-abelian and phaseless. Results are computer-verified on two surfaces, the cubic ($m_D = 3, m_P = 2$) and the keystone ($m_D = m_P = 2$); the keystone was chosen as an adversarial falsification test and passed.

Contents

1 Introduction

A directive-based product (DBP) presents an algebraic relation $D \oplus S = P$ as a constraint surface $F(D,S,P) = 0$ together with an assignment of *roles*: D is the directive (the active operator), S the substrate (the passive operand), and P the product. The framework's defining intuition is an asymmetry: D acting on S yields P, whereas "S acting on D" is a category error. As an *algebraic* object, however, $F = 0$ is symmetric—one may solve for any variable—so the directional restriction appears to dissolve. The first contribution of this paper is to show that it does not dissolve; it is *re-encoded in the geometry*, as an asymmetry in the degree and monodromy of the inverse problems.

The natural way to probe this is composition. We compose a DBP surface with itself by the *forward glue* $D_2 := P_1$: the product of stage one becomes the directive of stage two (output feeds next operator). Eliminating the shared variable produces a single hypersurface $\widetilde{F} = 0$, and we ask what survives of the role structure in its branched-cover geometry.

Summary of results and certainty tiers. Throughout we tag each statement by how strongly it is established: [proven] (an exact algebraic fact), [symbolic, all m_D] (verified symbolically for the whole family $m_D = 2, 3, 4, \ldots$ at $m_P = 2$ but not proved in closed form for all m_D), [computer-verified] (established by numerically certified monodromy on a specific surface, with the gates of §A), [conjecture] (a conjecture, with the supporting evidence stated), and [interpretation] (an interpretive reading, not a theorem).

1. **No single Galois group; per-cover wreaths.** Each role-output cover of $\widetilde{F}$ has its own monodromy group, and each is a full wreath product imprimitive with respect to the seam (§4–5). [computer-verified]/[symbolic, all m_D]

2. **Degree spectrum law.** For $F = D^{m_D} + DS + P^{m_P} - 3$ at $m_P = 2$, the spectrum of (D_1, S_1, S_2, P_2) in $\widetilde{F}$ is (m_D^2, m_D, m_P, m_P^2). [symbolic, all m_D]

3. **Product cover** $\cong \mathrm{Sym}(m_P) \wr \mathrm{Sym}(m_P)$, directive cover $\cong \mathrm{Sym}(m_D) \wr \mathrm{Sym}(m_D)$ at $m_P = 2$; the seam is the block system in both. Cubic: D_4 (product) and $S_3 \wr S_3$ of order 1296 (directive). Keystone: D_4 on both. [computer-verified]

4. D_4 **equals the seam-partition stabiliser.** For the product cover, the monodromy group is exactly the stabiliser in S_4 of the seam-aligned partition $\{\{D_1, S_1\}, \{S_2, P_2\}\}$; three independent derivations agree (§4). [computer-verified]

5. **The omega substrate.** S_2 controls the downstream product-branch position with the signature $\partial A/\partial S_2 = 0$, $\partial C/\partial S_2 = 1$; loops in S_2 steer exactly one product pair (§6). [computer-verified]

6. **Recovery asymmetry.** Over n stages, root-directive recovery is degree m_D^n and final-product production is degree m_P^n; for the cubic 3^n vs 2^n (§7). [symbolic, all m_D]

7. **Abelian boundary on the phase reading.** The abelianisation of $S_3 \wr S_3$ is $(\mathbb{Z}/2)^2$; its $U(1)$ characters take values in $\{\pm 1\}$, and the cube-root structure lives in the non-abelian part with no $U(1)$ character (§8). [computer-verified]

2 The self-glue construction

Definition 1 (DBP surface, worked family)**.** We work with

$$F(D, S, P) \;=\; D^{m_D} + D\,S + P^{m_P} - 3 \;=\; 0,$$

with integers $m_D, m_P \geq 2$, role assignment $(D, S, P) = (\text{directive}, \text{substrate}, \text{product})$. The two running examples are the *cubic* ($m_D = 3, m_P = 2$, i.e. $D^3 + DS + P^2 - 3$) and the *keystone* ($m_D = m_P = 2$, i.e. $D^2 + DS + P^2 - 3$).

Construction 1 (Forward self-glue)**.** Take two copies of F. Stage one is $F(D_1, S_1, P_1) = 0$; stage two is $F(D_2, S_2, P_2) = 0$ with the *seam identification* $D_2 := P_1$. Writing

$$A \;:=\; 3 - D_1^{m_D} - D_1 S_1, \qquad \text{so that} \qquad P_1^{m_P} = A,$$

stage two becomes $P_1^{m_D} + P_1 S_2 + P_2^{m_P} - 3 = 0$. Eliminating the seam variable P_1 (by resultant with $P_1^{m_P} - A$) yields a four-variable hypersurface

$$\widetilde{F}(D_1, S_1, S_2, P_2) = 0.$$

At $m_P = 2$ the seam relation is $P_1^2 = A$ and the elimination is explicit. For the cubic, with $C := A + S_2$ and $B := 3 - P_2^2$, stage two reads $P_1 C = B$, so $P_1^2 = A$ gives

$$\boxed{\widetilde{F}_{\text{cubic}} = B^2 - A\,C^2} \qquad (A = 3 - D_1^3 - D_1 S_1,\ C = A + S_2,\ B = 3 - P_2^2).$$

For the keystone, $P_1^2 = A$ makes the directive term $D_2^2 = P_1^2 = A$ a constant in P_1, so stage two reads $P_1 S_2 = 3 - A - P_2^2$ and hence

$$\boxed{\widetilde{F}_{\text{key}} = (3 - A - P_2^2)^2 - A\,S_2^2} \qquad (A = 3 - D_1^2 - D_1 S_1).$$

Definition 2 (Lifted correspondence)**.** Rather than work on $\widetilde{F}$ alone we retain the seam P_1 as an explicit fibre coordinate. At $m_P = 2$ the lifted system is the tower of two quadratics

$$P_1^2 = A, \qquad P_2^2 = 3 - P_1\,\text{(stage-two coupling)},$$

which for the cubic is $P_2^2 = 3 - P_1 C$ and for the keystone $P_2^2 = 3 - P_1(P_1 + S_2)$. The seam is the intermediate sheet over which the product sits.

3 Role-output covers and the degree spectrum

$\widetilde{F} = 0$ is a three-dimensional hypersurface in four variables; fixing which three variables form the base and solving for the fourth gives a finite branched cover. We call these the *role-output covers.*

Proposition 1 (No single Galois group)**.** *The monodromy group of $\widetilde{F}$ depends on the choice of output role. Concretely, the directive cover (solve for D_1) and the product cover (solve for P_2) have different degrees and different groups. A naive "*$\mathrm{Gal}(\widetilde{F}/base)$*" over a base in which the seam variable A is treated as a free parameter is the group of a* decoupled *problem and does not describe any actual cover.*

Remark 1. Proposition 1 is a correction of a natural error: one is tempted to assign $\widetilde{F}$ a single group (e.g. a product $S_3 \times D_4$ obtained by treating A as free). This fails on a counting obstruction—the directive cover is transitive of degree 9, so 9 must divide its order, but $|S_3 \times D_4| = 48$ is not divisible by 9. The honest object is the per-output-cover group.

Computational Result 1 (Degree spectrum law, $m_P = 2$). For $F = D^{m_D} + DS + P^2 - 3$ the degrees of (D_1, S_1, S_2, P_2) in $\widetilde{F}$ are

$$(\deg_{D_1}, \deg_{S_1}, \deg_{S_2}, \deg_{P_2}) \;=\; (m_D^2,\ m_D,\ m_P,\ m_P^2),$$

verified symbolically for $m_D \in \{2, 3, 4\}$. [symbolic, all m_D] The directive and its own substrate carry m_D-powers; the product and its substrate (the omega role, §6) carry m_P-powers.

surface	$\deg_{D_1}$	$\deg_{S_1}$	$\deg_{S_2}$	$\deg_{P_2}$
cubic (m_D=3, m_P=2)	9	3	2	4
keystone (m_D=2, m_P=2)	4	2	2	4
generic m_D (at m_P=2)	m_D^2	m_D	2	4

The keystone spectrum $(4, 2, 2, 4)$ is symmetric under $D_1 \leftrightarrow P_2$ precisely because $m_D = m_P$; the cubic cannot be, because $m_D \neq m_P$. This $D_1 \leftrightarrow P_2$ symmetry is the algebraic shadow of the directive/product duality.

4 The product cover

Solving $\widetilde{F}$ for P_2 over (D_1, S_1, S_2): A is single-valued, the seam P_1 has m_P values ($P_1^{m_P} = A$), and the product P_2 has m_P values. Both layers are m_P-fold extractions, so the cover factors as m_P seam blocks of m_P sheets each.

Computational Result 2 (Product-cover monodromy). The product cover is the full wreath $\mathrm{Sym}(m_P) \wr \mathrm{Sym}(m_P)$, of degree m_P^2, imprimitive with the seam ($\pm P_1$ for $m_P = 2$) as block system. [computer-verified] at $m_P = 2$. For both the cubic and the keystone this group is

$$\mathrm{Sym}(2) \wr \mathrm{Sym}(2) \;\cong\; D_4 \quad \text{(order 8)}.$$

The measurement uses three based loops from the basepoint $A_0 = 1$, $C_0 = 1$ (cubic) on the four lifted sheets 0=$(+P_1, +P_2)$, 1=$(+P_1, -P_2)$, 2=$(-P_1, +P_2)$, 3=$(-P_1, -P_2)$. With residual gates on *both* equations ($\leq 4 \times 10^{-15}$), bijective closure, and motion-ratio safety:

$$\tau_A = (0\,2)(1\,3) \text{ (seam)}, \quad \tau_+ = (0\,1) \ (+P_1 \text{ product}), \quad \tau_- = (2\,3) \ (-P_1 \text{ product}).$$

The conjugation $\tau_A \tau_+ \tau_A^{-1} = \tau_-$ holds, so the group is non-abelian; it contains the 4-cycle $(0\,2\,1\,3)$ and has order 8.

Computational Result 3 (D_4 is the seam-partition stabiliser). The product-cover group equals the stabiliser in S_4 of the seam-aligned partition $\{\{D_1, S_1\}, \{S_2, P_2\}\}$ (equivalently $\{\{0, 1\}, \{2, 3\}\}$ on the lifted sheets), the order-8 "V_4-core". [computer-verified]

Three independent derivations agree on this order-8 group: (i) the degree spectrum $(9, 3, 2, 4)$ is faithful with four distinct degrees, ruling out a transitive S_4; (ii) the seam-lock gradient identity $\partial\widetilde{F}/\partial D_1 : \partial\widetilde{F}/\partial S_1 = (3D_1^2 + S_1) : D_1$, the first-stage normal ratio, distinguishes the seam-aligned pairing; (iii) the lifted monodromy realises exactly the partition stabiliser. The group is dihedral, not the Klein four group, because the product sits *over* the seam ($P_2^2 = 3 - P_1 C$): circling the seam swaps which product branch point is encircled, which is the relation $\tau_+ \leftrightarrow \tau_-$.

5 The directive cover

Solving $\widetilde{F}$ for D_1 over (S_1, S_2, P_2): now P_2 (hence B) is fixed, so A is no longer single-valued—it satisfies the A-*block polynomial* obtained by eliminating D_1 from the stage-two relation. Each admissible A then yields m_D values of D_1 through $A = 3 - D_1^{m_D} - D_1 S_1$.

Computational Result 4 (A-block count, $m_P = 2$). At $m_P = 2$ the A-block polynomial has degree m_D in A, verified symbolically for $m_D \in \{2, 3, 4\}$. [symbolic, all m_D] Hence the directive cover factors as m_D blocks (the A-values) of m_D sheets (the D_1-roots within each block), total degree m_D^2, in agreement with Result 1.

Computational Result 5 (Directive-cover monodromy). At $m_P = 2$ the directive cover is the full wreath $\mathrm{Sym}(m_D) \wr \mathrm{Sym}(m_D)$ of order $(m_D!)^{m_D}\, m_D!$, imprimitive with the A-values as block system. [computer-verified] at $m_D \in \{2, 3\}$:

$$\text{cubic: } S_3 \wr S_3 \text{ (order 1296)}, \qquad \text{keystone: } \mathrm{Sym}(2) \wr \mathrm{Sym}(2) \cong D_4 \text{ (order 8)}.$$

For the cubic, the 9 sheets split into 3 A-blocks of 3; nine "within-block" lassos realise S_3 in each block (the base S_3^3), and three "A-branch" lassos realise S_3 on the blocks. The generated group has order $1296 = 6^3 \cdot 6$, is transitive and imprimitive, with block kernel of order $216 = 6^3$ and block action S_3—the full wreath. For the keystone, the 4 sheets split into 2 A-blocks of 2; within-block lassos give $(0\,3), (1\,2)$ and a block-swap lasso (reached by looping P_2, not S_1) gives $(0\,1)(2\,3)$, generating D_4. The cover is connected because the A-block discriminant $\Delta = S_2^2(12 - 4P_2^2 + S_2^2)$ is not a square in $\mathbb{Q}(S_2, P_2)$, so the two A-values fuse.

Remark 2 (The directive blow-up). The contrast is the central asymmetry of the paper: when $m_D \neq m_P$, inverting toward the directive is vastly larger than producing the product. For the cubic the directive cover (1296) dwarfs the product cover (8).

6 The omega substrate

The second-stage substrate S_2 enters $\widetilde{F}$ only through the stage-two coupling and never through the seam. We make this quantitative.

Computational Result 6 (Omega signature). S_2 satisfies $\partial A/\partial S_2 = 0$ and (writing C for the stage-two directive) $\partial C/\partial S_2 = 1$: it moves the product directive one-to-one while leaving the seam invariant. No other coordinate does this— $\partial A/\partial S_1 = -D_1 \neq 0$ and $\partial A/\partial D_1 = -m_D D_1^{m_D-1} - S_1 \neq 0$. [proven]

Computational Result 7 (Branch authority). With A held fixed, the product branches sit at $C = \pm 3/\sqrt{A}$, i.e. at

$$S_2^{(+)} = +\tfrac{3}{\sqrt{A}} - A \quad \text{(the } +P_1 \text{ branch)}, \qquad S_2^{(-)} = -\tfrac{3}{\sqrt{A}} - A \quad \text{(the } -P_1 \text{ branch)}.$$

A loop of S_2 around $S_2^{(+)}$ swaps *only* the $+P_1$ product pair ($(0\,1)$); a loop around $S_2^{(-)}$ swaps *only* the $-P_1$ pair ($(2\,3)$). [computer-verified] (residuals $\sim 10^{-15}$).

Thus S_2 is an *omega role*: a downstream branch-position controller. It carries no seam structure and no directive structure; it sets where the second-stage product branch lands, independently for each seam sign, and steering it moves exactly the targeted product pair. Tying S_2 to another role removes this: S_2=0 slaves the product branch to the seam, S_2=S_1 and S_2=D_1 make the controlling knob co-move A (contaminating the seam), and S_2=const freezes it. This also explains the substrate asymmetry that runs through the construction: *S_1 builds the seam, S_2 positions the product*—first-stage substrate is upstream geometry, second-stage substrate is the composition environment.

7 Iterated composition and the recovery asymmetry

The glue iterates: stage three identifies $D_3 := P_2$, and so on.

Computational Result 8 (Recovery exponents). For the three-stage composite of the cubic, the degree spectrum of $(D_1, S_1, S_2, S_3, P_3)$ is $(27, 9, 6, 4, 8)$. [symbolic, all m_D] In general the root-directive recovery degree after n stages is m_D^n and the final-product production degree is m_P^n:

$$\text{cubic backward: } 3, 9, 27, \dots = 3^n; \qquad \text{cubic forward: } 2, 4, 8, \dots = 2^n.$$

Producing the product is monotonically cheaper than reconstructing the directive, and the gap widens by a factor m_D/m_P per stage. This is the information-theoretic residue of the original "category error": inverting toward the operator is not the same operation as producing the product run backwards—it is a strictly higher-degree, more-branched cover, and the cost compounds through every seam. No stage is ever cheaper than the previous: the seam never rejects.

8 The abelian boundary on a phase interpretation

Covering-space monodromy and wave phase share their formalism: π_1 of a punctured base mapping to automorphisms, and for cyclic (abelian) covers this is a character into $U(1)$, with a lasso generating a phase $e^{2\pi i/k}$ and a branch point as a phase singularity. It is therefore tempting to read the whole monodromy as "wave structure" and the exponents m_D^n, m_P^n as a frequency spectrum. The following bounds that reading.

Computational Result 9 (Abelianisation). For the directive cover of the cubic, $G = S_3 \wr S_3$ has $|G| = 1296$, $|[G, G]| = 324$, and abelianisation

$$G^{\text{ab}} \;\cong\; (\mathbb{Z}/2)^2 \quad \text{(Klein four)}.$$

Hence every $U(1)$ character of G takes values in $\{\pm 1\}$ (the parities of the within-block and block permutations); an order-3 element maps to $+1$ under every one of them. The cube-root phase $e^{2\pi i/3}$ is *not* a character value. [computer-verified]

The cube roots are real but live elsewhere: they are the eigenvalues $e^{\pm 2\pi i/3}$ of a 3-cycle inside the 2-dimensional irreducible representation, which does not factor through $U(1)$. Thus the phase reading is exact for the *abelian shadow*—the ± 1 sign-phases, the m_P-frequency—and categorically fails for the higher (cubic) structure, which is the non-abelian part carrying the directive and the backward cost. In particular, the forward/cheap structure is abelian and phase-like, while the backward/expensive structure is non-abelian and phaseless; the boundary between them is the boundary between $U(1)$ and what does not embed in it.

9 Two-surface evidence: the keystone as falsification test

The keystone $m_D = m_P = 2$ was chosen to be able to break the morphology: if the directive cover threw up an S_3 or any 3-cycle when both roles enter quadratically, the law "layer group = Sym(entry degree)" would die. It does not.

	cubic (m_D=3, m_P=2)	keystone (m_D=m_P=2)
degree spectrum (D_1, S_1, S_2, P_2)	$(9, 3, 2, 4)$	$(4, 2, 2, 4)$
product cover	D_4 (order 8)	D_4 (order 8)
directive cover	$S_3 \wr S_3$ (order 1296)	D_4 (order 8)
cube-root strata	yes (directive)	none
block structure	3×3 / 2×2	2×2 on both

Every entry matches the frozen prediction: both covers D_4, both degree 4, both imprimitive with 2 blocks of 2, and no 3-cycle anywhere (all branching is quadratic). The wreath-with-seam-imprimitivity structure now stands on two surfaces, the second adversarial.

10 Discussion and open problems

The seam as the fourth structural element. [interpretation] The seam—the directed identification $P_1 = D_2$—is the load-bearing object: it is the imprimitivity system of every cover, it carries the forward orientation of the composition, and eliminating it *is* enforcing that the product was produced and passed on. Read as direction, it orients composition forward; iterated, it orders the stages (an ordinal, not metric, arrow—there is no duration in the construction); read as confirmation, it certifies the product. We emphasise this is interpretation, and that the construction's confirmation is unconditional: the seam has no failure branch (Result 8), so "feedback" overreaches.

Beyond $m_P = 2$. [conjecture] The product cover is structurally $\mathrm{Sym}(m_P) \wr \mathrm{Sym}(m_P)$ for any m_P (two m_P-fold extractions), but this is tested only at $m_P = 2$. The decisive open test is $m_P = 3$: does the product side climb to $S_3 \wr S_3$ as the directive side did? At $m_P \geq 3$ the seam is a higher root and the directive-cover A-block degree is no longer a clean power of A; the directive-cover law $\mathrm{Sym}(m_D) \wr \mathrm{Sym}(m_D)$ is established (block structure) only for $m_P = 2$.

Fullness of the wreath. [conjecture] The block structure of both covers is symbolic for all m_D (at $m_P = 2$); that the monodromy realises the *full* wreath rather than a proper subgroup is computer-verified only at $m_D \in \{2, 3\}$. A proof that the lassos generate the full base and full block actions for all m_D would upgrade Results 2,5 to theorems.

Universality across DBP surfaces. All results are for the specific family $D^{m_D} + DS + P^{m_P} - 3$. Whether the seam-imprimitive wreath structure holds for general separable DBP surfaces $h(P) = k(D, S)$ is open; separability of the product is likely a genuine hypothesis (non-separable cross terms break the clean radical tower).

A Computational protocol

Monodromy groups are obtained by numerical analytic continuation of the sheets around based loops in the punctured base, under the following gates, all of which must pass at every accepted continuation step:

1. **Root residual** on *every* defining equation (e.g. $|P_1^2 - A|$ and $|P_2^2 - (3 - P_1 C)|$), required $< 10^{-9}$ (achieved $\leq 4 \times 10^{-15}$).
2. **Bijective closure**: the nearest-neighbour sheet assignment must be a bijection.

3. **Motion-ratio safety**: max sheet motion over min sheet separation is tracked and kept small ($\sim 10^{-5}$).
4. **Group from permutations**: the group is generated from the measured permutations and its order computed by a permutation-group routine, never inferred.
5. **Byte-stability**: each run is repeated and the report's content hash (timestamp excluded) compared; runs are byte-identical.

Symbolic results (degree spectra, A-block degrees, the omega derivatives, the abelianisation) are exact computer algebra. The product-cover artifact (loops, generators, gates, generated group, partition-stabiliser check, and a test that re-derives D_4 from the measured permutations) is archived separately.

Acknowledgements

Computational consultation and verification were carried out in an interactive session; all stated groups were re-derived from measured permutations under the gates of Appendix A. *[To be completed.]*